4.6 The great tree of life

Science inquiry skills

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In his book *On the Origin of Species*, Charles Darwin drew a diagram he called the 'Great Tree of Life'. The 'tree' diagram showed how species should be related to each other if this theory of evolution was correct. Each branching of the tree showed where new species evolved because new structures appeared in the fossil record of the descendants.

phylogeny (*n*) a diagram showing the evolutionary relationships between species

Phylogeny

Darwin's technique is still used today. Often hundreds of features are used to create the 'tree'. Most of these are based on anatomy. However, now biologists also have molecular evidence such as DNA structure to help them. Consider Figure 4.6.1 which shows how tetrapods may have evolved. This diagram shows the branching tree structure originally developed by Darwin. This type of diagram is now called a phylogeny.

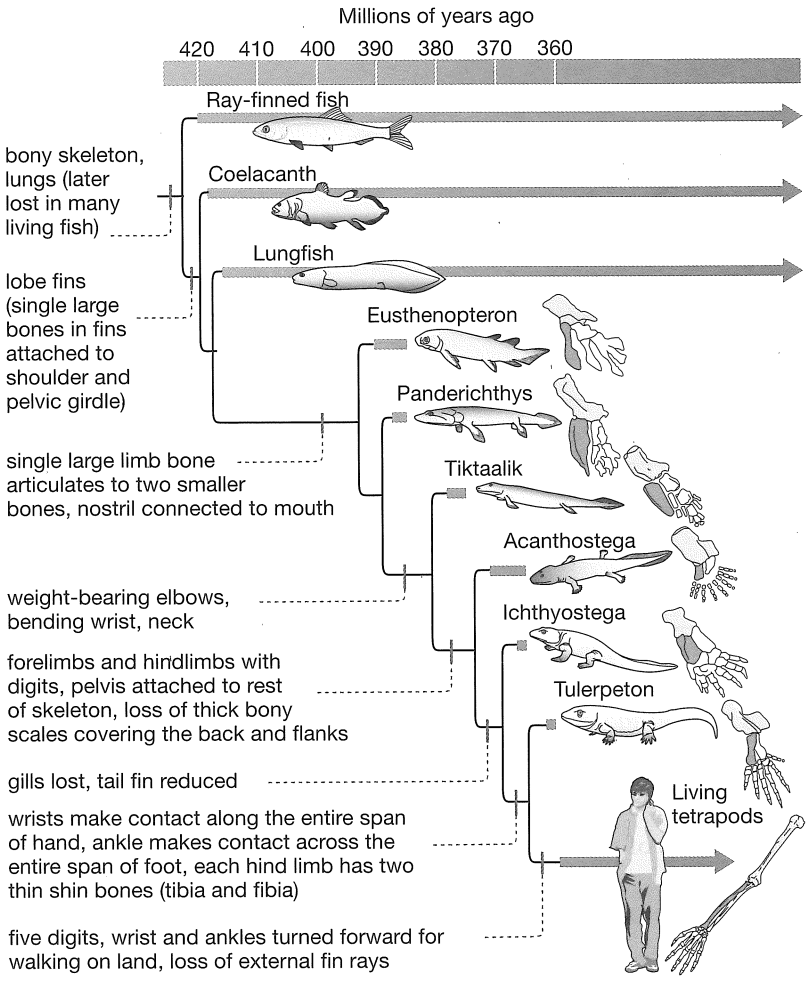


Figure 4.6.1 Phylogeny of tetrapods

Reading a phylogeny diagram

To read the diagram, start at the far left-hand side where the label says 'Bony skeleton, lungs'. You proceed through the diagram like you would use a biological key. The first line branches into two choices: 'ray-finned fish' and another branch that is labelled (with a small vertical line) 'lobe fins'. What the diagram shows is that the ray finned fish and the lobe-finned fish had a common ancestor. That ancestor had a bony skeleton and structures similar to lungs.